

Identifying Exoplanets in Habitable Zones

Isabella Love



Project Goal

- Model which exoplanets fall inside their stars' habitable zones
- Identify/model the relationship between stellar temperature and luminosity and HZ distance
 - HZ = region where surface liquid water could exist

Methods Overview

- Loaded exoplanet dataset into pandas

Removed rows that did not include:

- Stellar luminosity
- Stellar temperature
- Planet semi-major axis

Computed HZ boundaries using stellar luminosity

HZ boundaries:

$$r_{inner} = \sqrt{\frac{L}{s_{inner}}} \quad r_{outer} = \sqrt{\frac{L}{s_{outer}}}$$

- L = stellar luminosity in solar units
- $s_{inner} = 1.1$ (conservative runaway greenhouse limit)
- $s_{outer} = 0.53$ (maximum greenhouse limit)

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Planetary Systems

Planet Name	Host Name	Default Parameter Set	Number of Stars	Number of Planets	Discovery Method	Discovery Year	Discovery Facility	Solution Type	Controversial Flag	Planetary Parameter Reference	Orbital Pe
✓ AU Mic b	AU Mic	1	1	4	Transit	2020	Transiting Exoplanet Published Confirmed	0	Witrock et al. 2023	6.46308e+00	
✓ AU Mic c	AU Mic	1	1	4	Transit	2021	Transiting Exoplanet Published Confirmed	0	Witrock et al. 2023	18.8999e+00	
✓ BD+05 4868 A b	BD+05 4868 A	1	2	1	Transit	2025	Transiting Exoplanet Published Confirmed	0	Hon et al. 2025	1.27186e+00	
✓ BD-14 3065 b	BD-14 3065 A	1	3	1	Transit	2024	Transiting Exoplanet Published Confirmed	0	Subjak et al. 2024	4.28897e+00	
✓ DS Tuc A b	DS Tuc A	1	2	1	Transit	2019	Transiting Exoplanet Published Confirmed	0	Newton et al. 2019	8.13826e+00	
✓ GJ 1252 b	GJ 1252	1	1	1	Transit	2020	Transiting Exoplanet Published Confirmed	0	Luque & Palé 2022	0.51823e+00	
✓ GJ 143 b	GJ 143	1	1	2	Transit	2019	Transiting Exoplanet Published Confirmed	0	Dragoni et al. 2019	36.4253e+00	
✓ GJ 228 b	GJ 228	1	1	1	Transit	2024	Transiting Exoplanet Published Confirmed	0	Tay et al. 2024	1.74646e+00	
✓ GJ 3090 b	GJ 3090	1	1	1	Transit	2022	Transiting Exoplanet Published Confirmed	0	Almeida et al. 2022	2.85310e+00	
✓ GJ 341 b	GJ 341	1	1	1	Transit	2025	Transiting Exoplanet Published Confirmed	0	DiTomasso et al. 2025	7.57686e+00	
✓ GJ 3473 b	GJ 3473	1	2	2	Transit	2020	Transiting Exoplanet Published Confirmed	0	Kemmer et al. 2020	1.19800e+00	
✓ GJ 357 b	GJ 357	1	1	3	Transit	2019	Transiting Exoplanet Published Confirmed	0	Odio et al. 2023	3.93060e+00	
✓ GJ 367 b	GJ 367	1	1	3	Transit	2021	Transiting Exoplanet Published Confirmed	0	Goffin et al. 2023	0.32492e+00	
✓ GJ 3629 b	GJ 3629	1	1	2	Transit	2022	Transiting Exoplanet Published Confirmed	0	Bouvier et al. 2022	2.81623e+00	
✓ GJ 806 b	GJ 806	1	1	2	Transit	2023	Transiting Exoplanet Published Confirmed	0	Palte et al. 2023	0.92632e+00	
✓ Gliese 12 b	Gliese 12	1	1	1	Transit	2024	Transiting Exoplanet Published Confirmed	0	Kuzuhara et al. 2024	12.7614e+00	
✓ HD 101581 b	HP 56998	1	1	2	Transit	2025	Transiting Exoplanet Published Confirmed	0	Kurimoto et al. 2025	4.46569e+00	
✓ HD 101581 c	HP 56998	1	1	2	Transit	2025	Transiting Exoplanet Published Confirmed	0	Kurimoto et al. 2025	6.20401e+00	
✓ HD 108236 b	HD 108236	1	1	5	Transit	2020	Transiting Exoplanet Published Confirmed	0	Bonfanti et al. 2021	3.79596e+00	
✓ HD 108236 c	HD 108236	1	1	5	Transit	2020	Transiting Exoplanet Published Confirmed	0	Bonfanti et al. 2021	6.20244e+00	
✓ HD 108236 d	HD 108236	1	1	5	Transit	2020	Transiting Exoplanet Published Confirmed	0	Bonfanti et al. 2021	14.1756e+00	
✓ HD 108236 e	HD 108236	1	1	5	Transit	2020	Transiting Exoplanet Published Confirmed	0	Bonfanti et al. 2021	19.59002e+00	

Showing records 1 to 16 of 710 (26119 total) DOI:10.26133/NEA12

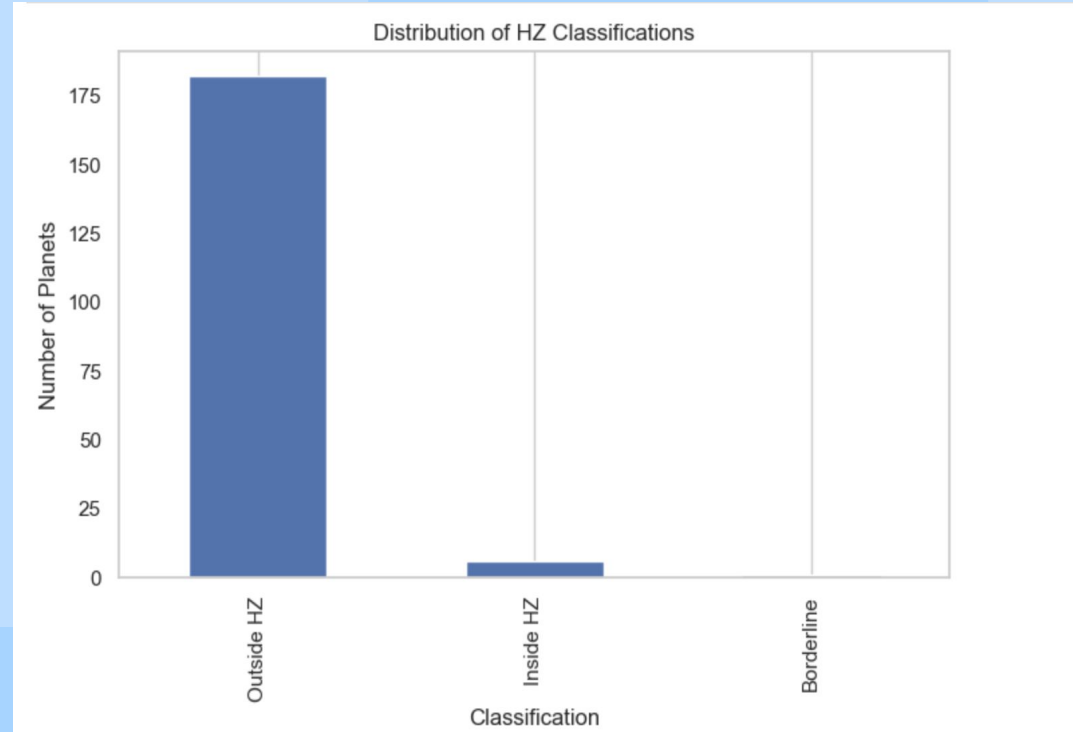
Clear Checked Clear All Reset Filters

Planet Classification

Inside HZ: semi-major axis between the HZ inner and outer radius

Borderline: within $\pm 10\%$ of either boundary

Outside HZ: everywhere else



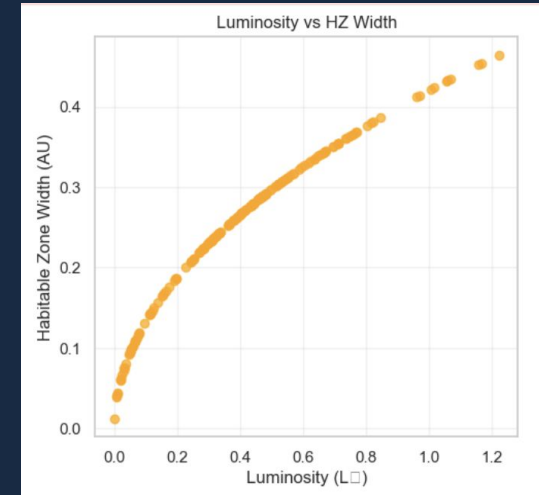
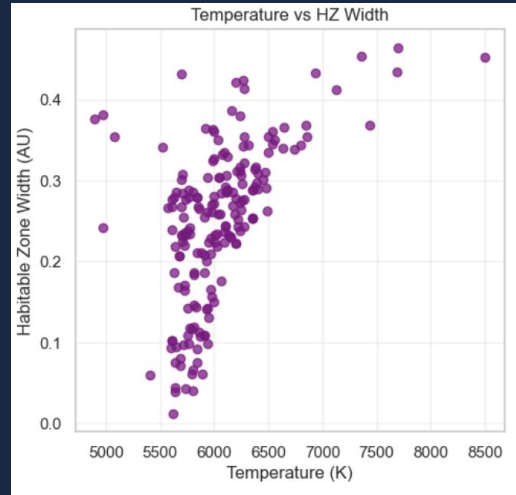
Stellar Properties & HZ Distance

Higher luminosity → HZ farther from star

Hotter stars → wide, distant HZ

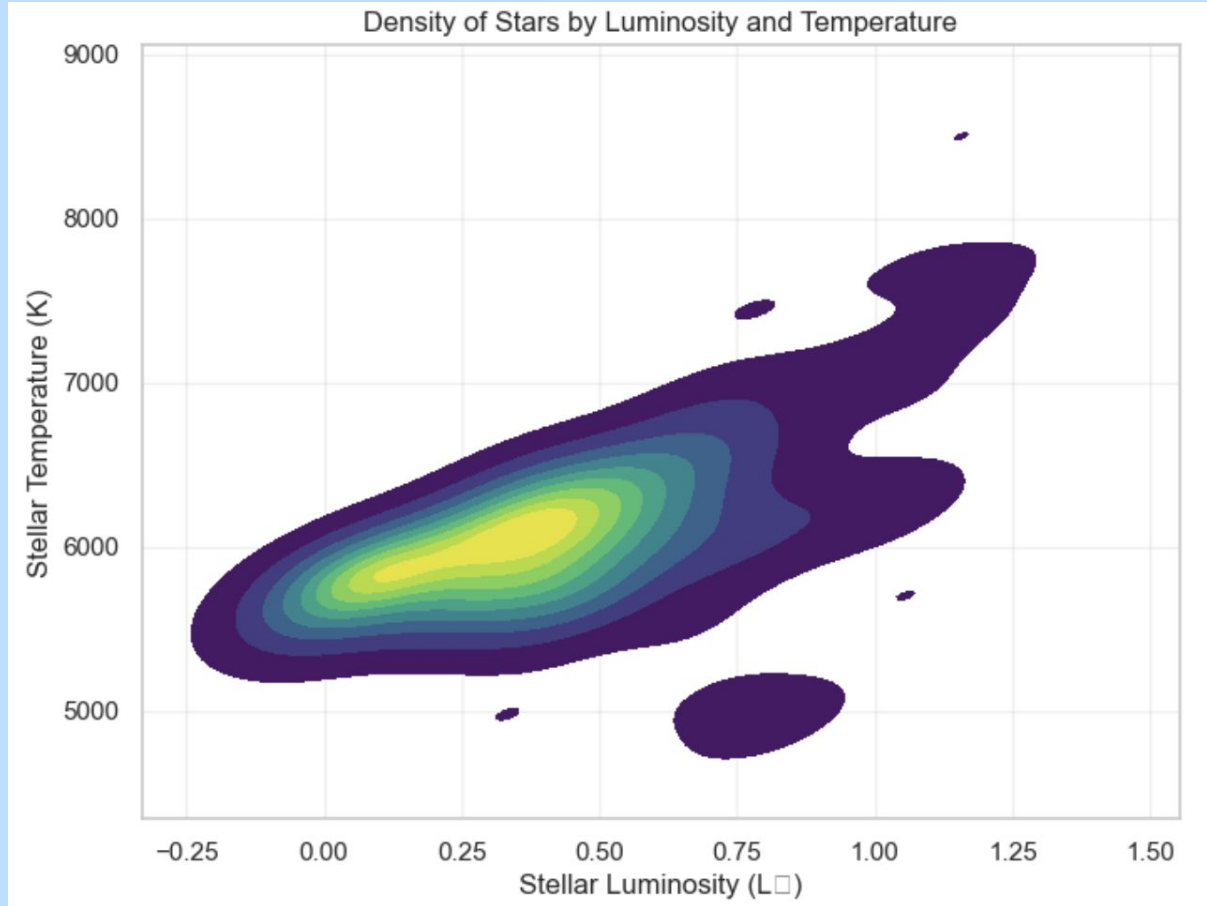
Cooler stars → narrow, close-in HZ

- Results match theoretical expectations



Density of Stellar Types

- Heatmap shows distribution of stars in luminosity-temperature space
- Dense/bright region = main-sequence stars
- Explains why most HZs fall in certain ranges
- Helps contextualize HZ distances



HZ Planet Candidates

- Successfully identified HZ planets using real NASA data
- HZ distance strongly depends on stellar luminosity and temperature
- Only a minority of known exoplanets lie inside the HZ

	pl_name	a	HZ_inner	HZ_outer
124	TIC 241249530 b	0.6410	0.609202	0.877647
126	TIC 279401253 b	0.3690	0.296816	0.427609
132	TIC 4672985 b	0.3300	0.321841	0.463661
357	TOI-2180 b	0.8280	0.607169	0.874718
481	TOI-3856 b	0.0325	0.028123	0.040516
553	TOI-4914 b	0.0980	0.090604	0.130529



Thank You!

Works Cited

- NASA. “NASA Exoplanet Archive.” Caltech.edu, 2019, exoplanetarchive.ipac.caltech.edu/.
- Kopparapu, R. K. et al. (2013). Habitable Zones Around Main-Sequence Stars. *Astrophysical Journal*.
- “The Habitable Zone.” Science.nasa.gov, NASA, 2024, science.nasa.gov/exoplanets/habitable-zone/.